Lecture 14

The toric code: topology protects

Objectives. *Lecture 13 ended with a question:* can protection grow with the size of the system? *The toric code answers it. Its qubits sit on the edges of a square lattice on a torus, each star and plaquette acts on four qubits, and it encodes two qubits (Theorem 14.2). Its distance is the size L (Theorem 14.3). Its Pauli errors are strings, with endpoint syndromes called anyons (Theorem 14.4). This is the second half of the third answer, and the door to Part II.*

By the end of this lecture you will be able to:

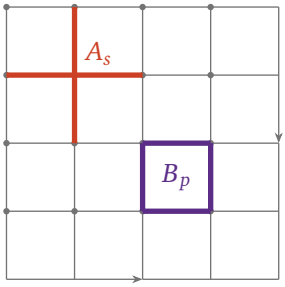
- Define the toric code, and prove that its stars and plaquettes commute and that it encodes $k = 2$.
- Show that contractible loops and dual loops are products of stabilizers and that the straight loops around the torus are logical, and state that $d = L$.
- Derive the endpoint syndromes of Pauli strings, prove that taking an e around an m gives -1, and show that the logical X and Z anticommute for the same reason.
- Say what "protected" means: what least-weight decoding corrects on $L = 3$, and how failure falls as L grows, under a stated model.

14.1 The lattice, commutation

Take $L \geq 2$ and the $L \times L$ square lattice on a torus: vertices (i, j) with $i, j \in \mathbb{Z}_L$ (row i, column j), the horizontal edge $h(i, j)$ from (i, j) to $(i, j+1)$, and the vertical edge $v(i, j)$ from (i, j) to $(i+1, j)$, with all indices modulo L: the right side is glued to the left, the bottom to the top. There are L^2 vertices, L^2 faces and $2L^2$ edges. The face $p(i, j)$ has the edges $h(i, j)$, $h(i+1, j)$, $v(i, j)$ and $v(i, j+1)$. As $L \geq 2$, the four edges at a vertex are distinct, so are the four of a face, and each edge has two distinct ends and borders two distinct faces.

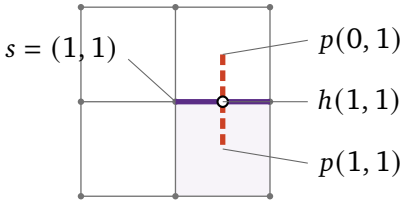
Definition 14.1 (The toric code)**.** Put a qubit on each of the $2L^2$ edges. The *star* of a vertex s is $A_s = \prod_{e \ni s} X_e$, over the four edges at s; the *plaquette* of a face p is $B_p = \prod_{e \in \partial p} Z_e$, over the four edges of its boundary. The *toric code* is the code $C(S)$ of the group S generated by all stars and plaquettes.

The figure marks the star at $(1,1)$ and plaquette $p(2,2)$ on $L = 4$. Arrows identify the sides that are glued.



Stars commute with stars, and plaquettes with plaquettes. A star and a plaquette share no edge unless s is a corner of p, and then exactly the two edges of ∂p at s: they differ on 0 or 2 qubits, so they commute (§12.1). So S is abelian; that $-\mathbb{1} \notin S$ is proved below. This is Kitaev's definition (1997, eq. (1), p. 2), and Dennis, Kitaev, Landahl and Preskill's (p. 4); Kitaev writes k, and Kitaev, Shen and Vyalyi write r (p. 170), for our L.

The enlarged patch distinguishes the lattice from its dual. Each solid edge carries one physical qubit, marked here by a small circle on $h(1,1)$. The dashed dual edge joins the centres of the two adjacent faces:



The vertex $s = (1,1)$ is an endpoint of the physical edge. The two faces are $p(0,1)$ and $p(1,1)$. A dual edge specifies the physical edge it crosses: applying X along that dual edge applies X to the qubit on $h(1,1)$. There is no additional qubit at a face centre.

A set c of edges is a *loop* if every vertex meets an even number of its edges, and a *dual loop* if every face has an even number of them on its boundary: the *dual edges*, one across each edge of c joining the two faces at it, then form closed paths through the faces. As the stars are products of X's and the plaquettes of Z's, a product $X(a)Z(b) = \prod_{e\in a} X_e \prod_{e\in b} Z_e$, times a phase, commutes with every star exactly when b is a loop, and with every plaquette exactly when a is a dual loop.

14.2 $k = 2$

Theorem 14.2 ($k = 2$). *For $L \geq 2$, a product of stars and plaquettes is $\pm\mathbb{1}$ only if it uses all the stars or none and all the plaquettes or none, and it is then $\mathbb{1}$; so $-\mathbb{1} \notin S$. Dropping one star and one plaquette leaves $2L^2 - 2$ independent generators of S, and the toric code has dimension 4: $k = 2$.*

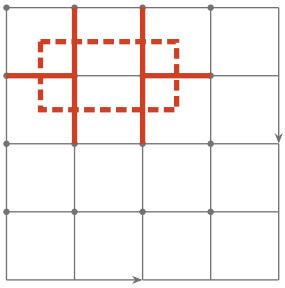
Proof. Let T be a set of vertices. Each edge lies in the stars of its two ends, and $X_e^2 = \mathbb{1}$, so $\prod_{s\in T} A_s = X(a)$, with a the set of edges that have exactly one end in T. If a is empty, every neighbour of a vertex of T is in T, and as the lattice is connected, T is empty or all of it. Likewise, for a set Q of faces, $\prod_{p\in Q} B_p = Z(b)$, with b the set of edges that border exactly one face of Q, and b is empty only if Q is empty or all of it, as the faces are connected across edges. Stars commute with plaquettes, so $\prod_T A_s \prod_Q B_p = X(a)Z(b)$, in any order and with no phase. It is a multiple of $\mathbb{1}$ only if a and b are empty: a qubit in a or b carries X, Z or

$XZ = -iY$. So the only products that are $\pm\mathbb{1}$ are the four with T and Q each empty or all, and each is $\mathbb{1}$.

Now drop A_{s_0} and B_{p_0}. If a remaining generator were a product of the others, a product of stars and plaquettes with $s_0 \notin T$, $p_0 \notin Q$ and T or Q nonempty would be $\mathbb{1}$, which we excluded. And the dropped two are products of the rest, $A_{s_0} = \prod_{s\neq s_0} A_s$ and $B_{p_0} = \prod_{p\neq p_0} B_p$, so the rest still generate S. By Theorem 12.3 the code has dimension $2^{2L^2-(2L^2-2)} = 4$. ∎

A simple closed lattice curve is called *contractible* when it bounds a disk on the torus, so it can be shrunk continuously to a point. The same definition applies to a simple curve in the dual lattice. Crossing the cut used to draw the torus as a square does not by itself make a curve non-contractible.

Two neighbouring stars illustrate cancellation of a shared edge:



The stars at $(1,1)$ and $(1,2)$ share $h(1,1)$, which cancels. Their product acts with X on the six edges crossed by the dashed dual loop. A contractible dual loop is a product of stars.

The two relations $\prod_s A_s = \prod_p B_p = \mathbb{1}$ make the $2L^2$ generators dependent, as were Lecture 12's Z_1Z_2, Z_2Z_3, Z_1Z_3: counting them all would give $k = 0$. The proof that there are no further relations uses the connectedness of the lattice and its faces.

14.3 Loops as logical operators; $d = L$

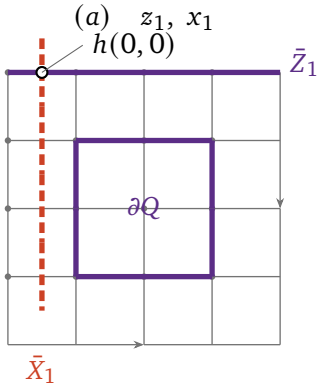
Theorem 14.3 (Logical operators · *(iii) cited*). *(i) Z on the boundary of a set of faces is the product of their plaquettes, and X on the edges leaving a set of vertices is the product of their stars: such loops act as $\mathbb{1}$ on the code.*

(ii) The loops $z_1 = \{h(0,j)\}_j$ and $z_2 = \{v(i,0)\}_i$ around the torus, and the dual loops $x_1 = \{h(i,0)\}_i$ and $x_2 = \{v(0,j)\}_j$, give $\bar{Z}_a = Z(z_a)$ and $\bar{X}_a = X(x_a)$ in $N(S) \setminus \langle i\mathbb{1}\rangle S$, of weight L, with $\bar{Z}_a\bar{X}_b = (-1)^{\delta_{ab}}\bar{X}_b\bar{Z}_a$.

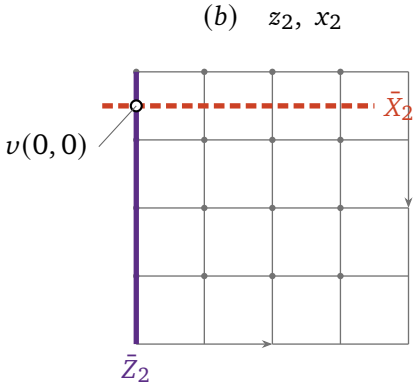
(iii) The distance is L.

Proof of (i) and (ii). (i) is in the proof of Theorem 14.2. (ii) Each vertex meets z_a in 0 or 2 edges, and each face meets x_a in 0 or 2, so the four commute with S. z_1 and x_1 share the edge $h(0,0)$, z_2 and x_2 the edge $v(0,0)$, and z_1, x_2 and z_2, x_1 share none. An element cg of $\langle i\mathbb{1}\rangle S$ commutes with every element of $N(S)$; $\bar{Z}_a$ anticommutes with $\bar{X}_a \in N(S)$, so it is not in $\langle i\mathbb{1}\rangle S$, nor is $\bar{X}_a$. ∎

The figure compares the boundary of four faces Q with the paired loops z_1 and x_1:



The operator $Z(\partial Q)$ is the product of the four plaquettes. Panel (a) marks the one physical edge $h(0,0)$ shared by z_1 and x_1. The second logical pair is shown separately:



Panel (b) marks the shared physical edge $v(0,0)$ of z_2 and x_2. A solid primal loop specifies the qubits carrying Z; a dashed dual loop specifies the crossed qubits carrying X. Each matched pair has one shared qubit, and therefore anticommutes. Here a pair is a loop and a dual loop that cross once. The unmatched pairs have no shared qubit and commute.

The boundary statement in Theorem 14.3 is more general than a statement about one simple contractible curve. For example, the boundary of a band of faces may consist of two parallel loops that each wind around the torus. Together they are a product of plaquettes. A loop that crosses x_1 an odd number of times, or x_2, cannot be the boundary of faces: it goes around the torus (loops modulo boundaries: $\mathbb{Z}_2$ homology, named here, not built).

Part (iii) is cited: every element of $N(S) \setminus \langle i\mathbb{1} \rangle S$ has a part, its X-string or its Z-string, that goes around the torus, and meets each of L disjoint rows or columns (Kitaev, p. 4; Dennis et al., pp. 5–6; Kitaev, Shen and Vyalyi, p. 171; Watrous, 2025, p. 456). Finite computer checks verify the distance on $L = 2$ and 3 by exhausting Pauli operators up to the claimed distance, and on $L = 4$ by checking their X and Z parts up to weight 4. These finite checks support the examples; the general statement remains the cited result.

14.4 Strings, anyons, -1

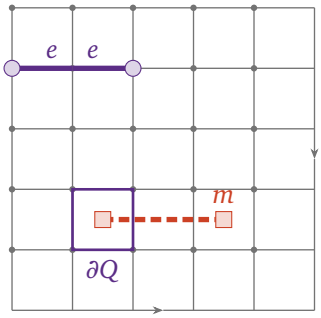
An error $X(a)$ violates the plaquettes p that meet a in an odd number of edges, and $Z(b)$ the stars that meet b in an odd number; a Y is both. Call a violated star an e and a violated plaquette an m. A *path* is a sequence of edges, each starting where the last ended; a *dual path* is a sequence of dual edges, from face to face.

Theorem 14.4 (Strings and anyons). *(i) Z on a path from a vertex u to a vertex $w \neq u$ violates exactly the stars at u and w; X on a dual path from a face f to a face $g \neq f$ violates exactly the plaquettes f and g.*

(ii) Let ψ be in the code, t a dual path from f to $g \neq f$, and Q a set of faces with $f \in Q$ and $g \notin Q$. Then $Z(\partial Q)\, X(t)\psi = -X(t)\psi$.

Proof. (i) Each passage of the path through a vertex uses two edges at it, and the two ends one each; an edge used twice cancels, as $Z_e^2 = \mathbb{1}$. So the star of a vertex meets the path in an odd number of edges exactly at u and w. The dual case is the same, faces for vertices. (ii) By Theorem 14.3(i), $Z(\partial Q) = \prod_{p \in Q} B_p$, and by (i), B_p anticommutes with $X(t)$ exactly for $p = f$ and $p = g$; one of them is in Q. So $Z(\partial Q)X(t) = -X(t)Z(\partial Q)$, and $Z(\partial Q)\psi = \psi$. ∎

The figure shows a primal Z path and a dual X path, with a loop around one end of the dual path:



A Pauli error is represented, up to phase, by a primal Z string and a dual X string. Its syndrome records their endpoints, counted modulo 2. Closed loops have no endpoint syndrome. In the figure, the solid primal path leaves two e's, and the dashed dual path two m's. Read (ii) as a process: an e pair is made by Z on one edge of ∂Q, one e is taken once along ∂Q, around the region Q that holds the m at f, and the pair is removed; the net operator is $Z(\partial Q)$, and the state changes sign. In the picture ∂Q crosses the m-string once. The logical $\bar{X}_a$ and $\bar{Z}_a$ anticommute for the same reason: their loops cross once. This is Kitaev's computation (§2, Fig. 4, p. 7; $W = -1$, p. 8); also Kitaev, Shen and Vyalyi (p. 175) and Dennis et al. (p. 9). Kitaev's "z-type particles" are our e's.

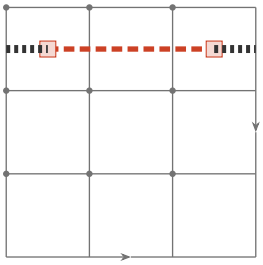
For this completed process the sign is a global phase: a reading of that state alone cannot see it (Theorem 3.7). If the loop process is coherently controlled, comparison with the branch without the loop makes the sign relative, as in Example 6.8.

The -1 is a monodromy: an e taken once around an m. It is not the phase of an exchange, as e and m are different particles. A phase other than $+1$ for one particle taken around another is what makes e and m anyons, which two space dimensions allow (*Symmetries*, L15, a starred reading).

14.5 $L = 3$ and suppression

Take $L = 3$: 18 qubits and $d = 3$. Look at X errors; Z errors are the same on the dual lattice. An error $X(a)$ shows the m's at the ends of its dual strings. *Least-weight decoding* applies an $X(b)$ of least weight with the same syndrome. Then $X(a)X(b) = X(a + b)$, with $a + b$ the edges in one of a, b but not both, has no syndrome: $a + b$ is a dual loop. The correction succeeds if $X(a + b)$ is a product of stars, which acts as $\mathbb{1}$ on the code, and otherwise leaves a logical error.

Every single X error is corrected: on $L = 3$ two faces share at most one edge, so the syndrome of one edge, the two faces at it, is that of no other single edge, and the decoder undoes the error itself. The figure shows two errors (dashed) and the least-weight correction (dotted) across the glued sides:



The errors X on $v(0, 1)$ and $v(0, 2)$ show m's at $p(0, 0)$ and $p(0, 2)$; $p(0, 1)$, which holds both, is not violated. The least-weight correction is the one edge between $p(0, 2)$ and $p(0, 0)$, across the glued sides: X on $v(0, 0)$ (dotted). The three make $X(x_2) = \bar{X}_2$, a logical error: the decoder closed the loop the short way round, and the error had gone the long way. With $d = 3$, one error is guaranteed to be corrected, not two. Of the 153 pairs of X errors, 18 fail this way, the three pairs on each of the six straight dual loops; 81 have one least-weight correction, which succeeds, and 54 have several, which differ by stars and all succeed: errors that differ by a star act alike on the code, which is degenerate for them (§11.6).

"Protected" does not mean that failures are absent; it means that they are suppressed as L grows, under a stated model of noise and decoding. Dennis et al. take independent X and Z errors on each qubit, each with probability p per round.

Perfect syndromes. A round of errors is corrected with a failure probability that tends to 0 as L grows when p is below about 10.5% with least-weight decoding, and below about 10.9% with optimal decoding (values of Kawashima and Aoki, and of Honecker et al., which they quote, pp. 17–18).

Faulty syndromes. When each syndrome bit is also wrong with probability p, independently across checks and rounds and of the data errors, they decode the history of T rounds of syndromes at once, by least weight and with exact classical processing, and prove that for $p < 1.14\%$, a lower bound on the threshold, the failure probability decreases exponentially with L, for T growing no faster than a polynomial in L (pp. 10–11, 18–20).

14.6 The third answer

Protection can grow with the size of the system. Each star and each plaquette of the toric code acts on four qubits, whatever L, while every logical operator goes around the torus: $d = L$. An error does harm only when its strings, with the decoder's, close a loop around the torus, and below threshold the probability of that falls exponentially with L, under the models above (Kitaev, p. 5; Dennis et al., p. 20). This completes the third answer: errors

are discrete and correctable (Lectures 10–11), stabilizer codes give them a working form (Lecture 12), a protected computation needs Clifford gadgets, magic states and a threshold (Lecture 13), and topology makes protection grow with size. The anyons here are abelian: taking e around m multiplies the state by a number.

Summary.

- *The toric code:* qubits on the $2L^2$ edges, commuting stars of X's and plaquettes of Z's; $k = 2$.
- *Loops:* contractible loops and dual loops are products of stabilizers; the straight loops around the torus are logical; $d = L$ (cited).
- *Anyons:* Pauli errors are strings with endpoint syndromes e or m; an e taken around an m gives -1, and $\bar{X}_a$, $\bar{Z}_a$ anticommute, as their strings cross once.
- *Protection:* on $L = 3$ one error is corrected, two can make a logical error; below threshold, failure falls exponentially with L, under a stated model.
- *Next:* Part II: anyons whose braiding acts by matrices, not numbers, compute by braiding.

References

A. Yu. Kitaev, arXiv:quant-ph/9707021 (1997), pp. 2–8. E. Dennis, A. Kitaev, A. Landahl and J. Preskill, arXiv:quant-ph/0110143 (2001), pp. 4–6, 9–11 and 17–20. Kitaev, Shen and Vyalyi, pp. 170–175. Watrous (2025), pp. 446–458. M. F. de Resende, arXiv:1712.01258, p. 17.